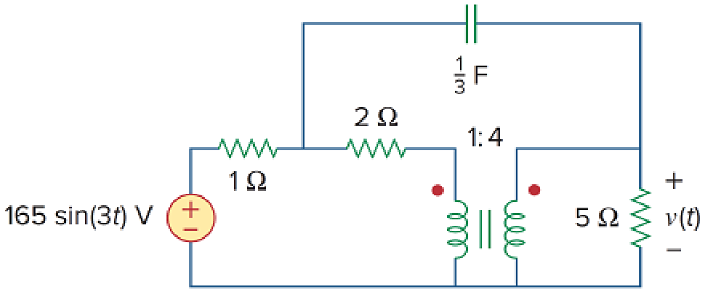
Chapter 13, Problem 47P

Problem

Find $v(t)$ for the circuit in Fig. 13.112.

Figure 13.112



Step-by-step solution

Step 1 of 5

Refer to Figure 13.112 in the textbook.

The expression for the input voltage source is,

$$\begin{aligned} v_s(t) &= 165\sin(3t) \\ &= 165\cos(3t - 90°) \end{aligned}$$

The phasor values are,

$$\mathbf{V}_s = 165\angle -90° \text{ V}$$
$$\omega = 3 \text{ rad/s}$$

The impedance of capacitor is,

$$\begin{aligned} \mathbf{Z}_C &= \frac{1}{j\omega C} \\ &= \frac{1}{j(3)\left(\frac{1}{3}\right)} \\ &= -j\ \Omega \end{aligned}$$

Step 2 of 5

Draw the circuit in frequency domain.

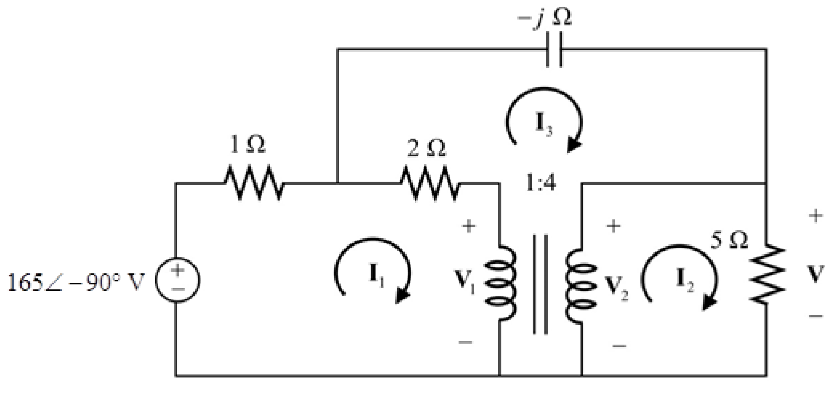


Figure 1

Step 3 of 5

From Figure 1,

$$\mathbf{V}_2 = \mathbf{V}$$
$$= 5\mathbf{I}_2$$

Calculate the primary and secondary voltage relation.

$$\frac{\mathbf{V}_1}{\mathbf{V}_2} = \frac{N_1}{N_2}$$
$$\frac{\mathbf{V}_1}{5\mathbf{I}_2} = \frac{1}{4}$$
$$\mathbf{V}_1 = \frac{5\mathbf{I}_2}{4}$$
$$\mathbf{V}_1 = 1.25\mathbf{I}_2$$

Step 4 of 5

Calculate the primary and secondary current relation.

$$\frac{\mathbf{I}_1}{\mathbf{I}_2} = \frac{N_2}{N_1}$$
$$\frac{\mathbf{I}_1}{\mathbf{I}_2} = \frac{4}{1}$$
$$\mathbf{I}_1 = 4\mathbf{I}_2$$

Apply Kirchhoff's voltage law to loop1.

$$-165\angle -90° + (1)\mathbf{I}_1 + (2)(\mathbf{I}_1 - \mathbf{I}_3) + \mathbf{V}_1 = 0$$
$$3\mathbf{I}_1 - 2\mathbf{I}_3 + \mathbf{V}_1 = 165\angle -90°$$

Substitute $1.25\mathbf{I}_2$ for $\mathbf{V}_1$ and $4\mathbf{I}_2$ for $\mathbf{I}_1$.

$$3(4\mathbf{I}_2) - 2\mathbf{I}_3 + 1.25\mathbf{I}_2 = 165\angle -90°$$
$$13.25\mathbf{I}_2 - 2\mathbf{I}_3 = 165\angle -90°$$
$$2\mathbf{I}_3 = 13.25\mathbf{I}_2 - 165\angle -90°$$
$$\mathbf{I}_3 = 6.625\mathbf{I}_2 - 82.5\angle -90°$$

Apply Kirchhoff's voltage law to outer loop.

$$(-j)\mathbf{I}_3 + \mathbf{V}_2 - 165\angle -90° + (1)\mathbf{I}_1 = 0$$
$$(-j)\mathbf{I}_3 + \mathbf{I}_1 + \mathbf{V}_2 = 165\angle -90°$$

Substitute $6.625\mathbf{I}_2 - 2$ for $\mathbf{I}_3$, $5\mathbf{I}_2$ for $\mathbf{V}_2$ and $4\mathbf{I}_2$ for $\mathbf{I}_1$.

$$(-j)(6.625\mathbf{I}_2 - 82.5\angle -90°) + 4\mathbf{I}_2 + 5\mathbf{I}_2 = 165\angle -90°$$
$$(-j)(6.625\mathbf{I}_2 + 82.5j) + 4\mathbf{I}_2 + 5\mathbf{I}_2 = -165j$$
$$(-j6.625)\mathbf{I}_2 + 82.5 + 9\mathbf{I}_2 = -165j$$
$$(9 - j6.625)\mathbf{I}_2 = -82.5 - j165$$

$$\mathbf{I}_2 = \frac{-82.5 - j165}{9 - j6.625}$$
$$= 2.8075 - j16.2667$$
$$= 16.5072\angle -80.2077° \text{ A}$$

Comments (1)

Anonymous

why does the KVL for loop 3 include the voltage source?

Step 5 of 5

Calculate the output voltage, $\mathbf{V}$.

$$\begin{aligned}\mathbf{V} &= \mathbf{V}_2 \\ &= 5\mathbf{I}_2 \\ &= 5(16.5072\angle -80.2077^\circ) \\ &= 82.536\angle -80.2077^\circ \text{ V}\end{aligned}$$

Convert the output voltage in phasor domain to time domain.

$$v(t) = 82.536\cos(3t - 80.2077^\circ) \text{ V}$$

Thus, the output voltage is $\boxed{82.536\cos(3t - 80.2077^\circ) \text{ V}}$.